**24** (a) Complete the table of values for $y = x^2 - x - 6$

| $x$ | –3 | –2 | –1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | 6 | 0 | -4 | – 6 | -6 | -4 | 0 |

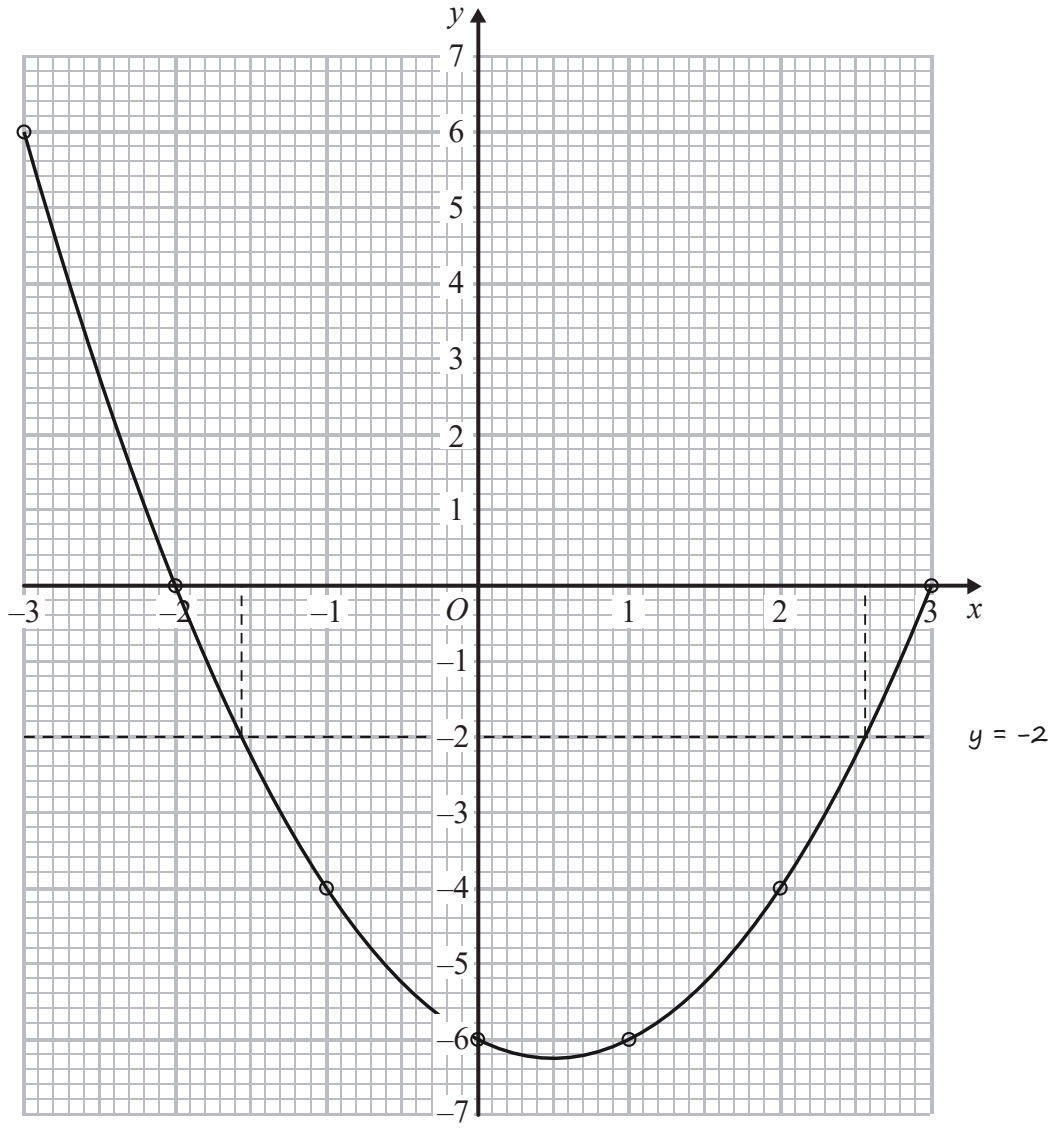
For example, when x = -2: $y = (-2)^2 - (-2) - 6 = 0$.

**2 marks total**
All five missing values correct

**(2)**

(b) On the grid, draw the graph of $y = x^2 - x - 6$ for values of $x$ from –3 to 3

**(2)**




**1 mark**
Plot at least five correct points

**1 mark**
Join with a smooth curve

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(c) Use your graph to find estimates of the solutions to the equation $x^2 - x - 6 = -2$

Draw y = -2 and read the x-coordinates where it meets the curve.

**1 mark**
Use the horizontal comparison line

**1 mark**
Both estimates correct (within reading tolerance)

-1.6 and 2.6

**(2)**

**(Total for Question 24 is 6 marks)**

**25** A force of 70 newtons acts on an area of 20 $cm^2$

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The force is increased by 10 newtons.
The area is increased by 10 $cm^2$

Helen says,

"The pressure decreases by less than 20%"

Is Helen correct?
You must show how you get your answer.

**(Total for Question 25 is 3 marks)**

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